Tour — Communication-Efficient Learning of Deep Networks from Decentralized Data

McMahan et al., 2017

Study repo: pleyva2004/communication-efficient-learning-of-deep-networks

A self-contained walking tour of this study. Read top-to-bottom for a complete first pass; jump by section if you already know the foundations. Estimated total time: **90 minutes**.

1. Reader's contract

By the end of this tour you will be able to:

- Derive why FedAvg cuts communication rounds and where non-IID data breaks it
- Reproduce the headline tradeoff and the Figure-1 barrier from runnable witnesses
- Map each proposed extension to the exact gap it closes (drift, bias, plateau, shared-init)

This tour assumes comfort with multivariable calculus / gradients (refresh: link) and basic probability / expectation (refresh: link).
Out of scope: differential privacy / secure aggregation; convergence theory for $E > 1$.

2. Foundations walk

Each row is one foundation node. Links resolve to the canonical concept page in the shared math-foundations repo.

Concept	Why it matters here	Link
expectation	defines the IID identity E[F$_k$] = $fandallunbiasednessarguments$	link
empirical risk	the finite-sum objective f is the empirical risk being minimized	link
partition of a set	the client decomposition f=sum$_k(n_k/n)F_kisexactbecauseP_kpartitionsthedata$	link
unbiased estimator	IID means each F$_kisanunbiasedproxyforf;Horvitz-Thompsonrestoresitundersampling$	link
gradient	FedSGD aggregates per-client gradients into grad f	link
gradient descent	FedSGD with C=1 is exactly full-batch GD	link
taylor theorem	the E>1 heterogeneity gap is a second-order Taylor remainder	link

Concept	Why it matters here	Link
hessian	the gap is a covariance of local Hessians $\mathrm{H}_k withgradients$	link
covariance	the drift term is exactly $\mathrm{Cov}_k(H_k, g_k)$	link
convex function	Jensen safety of averaging needs convex F_k	link
jensens inequality	gives f(avg) <= avg f for convex objectives	link
permutation group	hidden-unit relabelings are the symmetry behind the Fig-1 barrier	link
control variates	the SCAFFOLD fix subtracts a control variate to cancel drift	link
horvitz thompson estimator	the unbiased aggregation fix is inverse-probability weighting	link
linear assignment problem	permutation alignment solves an assignment problem	link
low rank factorization	federated LoRA communicates a low-rank adapter only	link

3. Paper concepts walk

The paper graph. Read in topological order; each row links to a Markdown concept page (prose plus formal definition) and a Python witness (CPU-runnable, under 30 seconds).

Concept	Role in the paper	Files
01. Finite-Sum Objective	the objective everything minimizes	`01-finite-sum-objective`
02. Client-Partition Decomposition	rewrites the objective per client (exact)	`02-client-partition-decomposition`
03. IID vs Non-IID	defines the regime FedAvg must survive	`03-iid-noniid-dichotomy`
04. FedSGD as Exact Gradient Descent	the FedSGD baseline = exact GD	`04-fedsgd-gradient-descent`
05. Gradient- vs Model-Averaging Equivalence	why model-averaging = gradient-averaging (1 step)	`05-gradient-model-averaging-equiva`
06. FedAvg: Iterated Local Steps	the FedAvg algorithm itself	`06-fedavg-local-iteration`
07. Local-Update Count u=nE/(KB)	the (C,E,B) compute-for-communication knob	`07-local-update-count`
08. Heterogeneity Gap = eta$^2 Cov_k(H_k, g_k)$	the leading E>1 deviation (client drift)	`08-heterogeneity-gap-covariance`
09. Client-Sampling Unbiasedness (Horvitz-Thompson)	justifies partial participation C<1	`09-client-sampling-unbiasedness`

Concept	Role in the paper	Files
10. Aggregation Erratum: normalize by m_t	the published aggregation erratum	`10-aggregation-erratum`
11. Parameter Averaging	Jensen	why averaging is safe when convex
`11-parameter-averaging-jensen`		
12. Shared-Init	the Permutation Barrier	why shared init is load-bearing
`12-shared-init-permutation-barrier`		

How the pieces fit. Start from the finite-sum objective (01) and its exact client decomposition (02); the IID/non-IID dichotomy (03) says how faithful each client's slice is. FedSGD (04) is exact GD, and the one-step gradient/model-averaging equivalence (05) is the hinge FedAvg (06) generalizes by iterating local steps — quantified by the update count (07). The two deep facts are the heterogeneity gap (08), which is exactly why non-IID hurts, and the sampling/erratum pair (09,10), which fix the aggregation. Finally (11,12) explain when averaging parameters is safe and why a shared per-round initialization is non-negotiable.

4. Improvements walk

Each proposal in `05-improvements.tex` gets exactly one validation file:

- **PROOF mode** — a standalone LaTeX proof under `proofs/` with theorem statement, complete proof, and discussion of where it would break.
- **MEASUREMENT mode** — an extended Python prototype under `improvements/` exposing `measure() -> dict` that prints a quantitative comparison table; deterministic under a fixed seed.

- **Control-Variate Drift Correction** — PROOF — `proofs/heterogeneity-control-variate.tex`
- **Unbiased Aggregation Estimators** — PROOF — `proofs/unbiased-aggregation.tex`
- **Adaptive Local-Work Schedule** — MEASUREMENT — `improvements/adaptive-local-work.py -> measure()`
- **Compute-Accounting Pareto (design)** — link-only — `design (E.2); see 05-improvements.tex`
- **Permutation-Aligned Averaging (Git Re-Basin for FL)** — PROOF — `proofs/permutation-aligned-av`
- **Federated LoRA (design)** — link-only — `design (T.2); see 05-improvements.tex`

5. What to do next

You have finished the tour. Pick one:

1. **Reproduce the sandbox claim.** `cd sandbox && pip install -r requirements.txt && python experiment.py`.

2. **Run a measurement.** Pick any improvement marked MEASUREMENT, run `python improvements/<slug>.` change one knob, re-run.
3. **Verify a proof.** Pick any improvement marked PROOF, audit the proof in `proofs/<slug>.tex`, then try to break it via the Discussion section.
4. **Form an opinion.** Fill in `03-opinions.md`.
5. **Extend the foundations graph.** Draft the missing concept at pleyva2004/math-foundations.